

Some basics of integration

It is not unusual to feel intimidated by integration. The purpose of these notes is to convey a simple message: Don't be afraid! Once you understand differentiation, integration is a small step. The treatment in these notes is somewhat informal, as the main aim is to make you comfortable with integration. We focus on functions of one variable.

1 Indefinite and definite integrals

To start on firm ground, consider some differentiable and univariate function $F(x)$. Differentiation to obtain the derivative, $F'(x)$, is a familiar process by now. In the following we will use the notation $f(x) = F'(x)$. The notation should remind you of the relationship between a cumulative distribution function (F) and a probability density function (f).

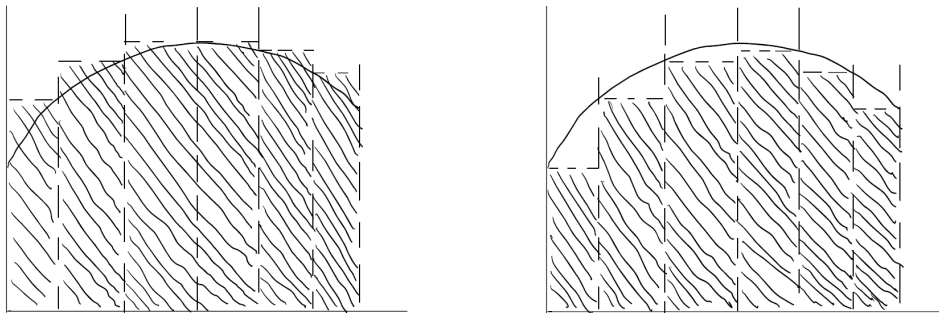
Imagine instead that you want to reverse the steps: To start with $f(x)$ and derive $F(x)$. More accurately, we want to find some function such that when we differentiate it, we will indeed get $f(x)$. This process is called anti-differentiation because we are going in the opposite direction of differentiation. There is, however, no unique solution to this problem. If $F(x)$ is an antiderivative, so is $F(x) + c$, where c is any constant. Hence, we call $F(x) + c$ an anti-derivative and call c the constant of integration. Nevertheless, terminology is often abused and we often refer to *the* anti-derivative because for most purposes the constant c will not matter (more on this later).

In terms of notation, we write

$$\int f(x)dx = F(x) + c,$$

to say that we are looking for the anti-derivative. The anti-derivative is also called the indefinite integral. The notation is explained on the next page.

In economics, we more commonly make use of *definite integrals*. Imagine that you have some arbitrary continuous function, $f(x)$, and that you want to calculate the area under the curve of $f(x)$ on some compact interval $[a, b]$. There are several ways to do so, giving rise to slightly different notions of integration (of varying levels of generality). The simplest method is Riemann integration. It approximates the area under the curve by first breaking $[a, b]$ into smaller chunks and then approximating $f(x)$ on each sub-interval. However, for any given partition of $[a, b]$, there are several ways to approximate $f(x)$. The figures on the next page illustrate two extreme ways, where you always over-estimate or under-estimate $f(x)$, respectively.



We are basically calculating the area of a number of rectangles and then summing them up. However, the height of each rectangle is ambiguous because there are (as we have seen) several ways to measure the height. However, in any case, with n sub-intervals, we can write our estimate as

$$\sum_{i=1}^n f(x_i) \Delta_i, \quad (1)$$

where Δ_i is the length of the i th interval and x_i is the point at which f is evaluated in the i th interval. We are not going to be using (1) directly. Rather, it serves the purpose of explaining where the following notation comes from, thus hopefully de-mystifying it.

As the partition gets finer and finer, you can see that the two ways of estimating the area converge to each other. The sub-intervals Δ_i get smaller and smaller and the height of the rectangles in either case converge to the same value, $f(x_i)$ (as the interval gets smaller, x_i gets “squeezed” and is no longer ambiguous). This convergent value is our definite integral, which we write as

$$\int_a^b f(x) dx. \quad (2)$$

Comparing (1) and (2), you can roughly speaking think of $\int$ as meaning “summation” and of dx as an “infinitesimal small” change in x . We usually just refer to this as the (Riemann) integral, rather than the definite integral. In fact, the integral exists whenever $f(x)$ is bounded and continuous “almost everywhere” and $[a, b]$ is a compact (closed and bounded) interval. Under some conditions, integration is also possible when the domain is unbounded. Sometimes, you will see the notation

$$\int_a^b f(x) dx = \int_a^b dF(x),$$

which makes sense since $f(x) = \frac{dF(x)}{dx}$ or $dF(x) = f(x) dx$.

The *fundamental theorem of calculus* says that for *any* antiderivative $F(x) + c$,

$$\int_a^b f(x)dx = (F(b) + c) - (F(a) + c) = F(b) - F(a),$$

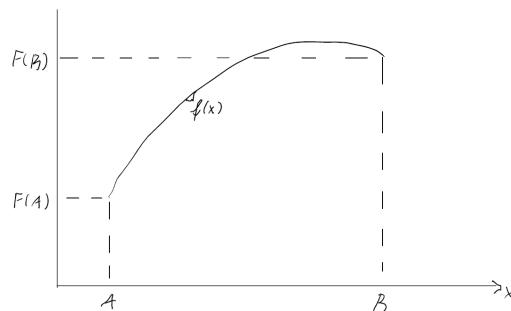
so c does not matter which is why $F(x)$ is sometimes just called *the* antiderivative. We use this result when we derive the equivalent and compensating variations (EV and CV) for welfare evaluations (with F =expenditure function and where f =Hicksian demand is the derivative of the expenditure function).

2 Discussion

First, in economics, we often use the implication that

$$F(b) = F(a) + \int_a^b f(x)dx,$$

which has a nice geometric interpretation that is useful for understanding the intuition:



In the figure, we start at $F(a)$. Each infinitesimal small step to the right increases the elevation by $f(x)$. After all the steps from a to b , we end up at $F(b)$. Thus, knowing where you start ($F(a)$) and the slope of each step ($f(x)$) is enough to infer where you end up ($F(b)$). An implication is that a firm's cost function can be inferred from knowing its fixed costs and its marginal cost function.

Second, this is a good place to clarify what we mean when talk about an increasing function of one variable. We say that $F(x)$ is *strictly increasing* if when we compare the function at any two distinct x values, say a and b , it holds that

$$b > a \implies F(b) > F(a).$$

If the last strict inequality is a weak inequality, $\geq$, then, depending on the source, the function may be referred to as weakly increasing or non-decreasing. Unfortunately,

if the term increasing is used without the strictly or weakly modifier, then it can be hard to infer what is meant, but it usually means weakly increasing.

Let us now specialize further by assuming that the function $F(x)$ is continuously differentiable, with derivative $F'(x) = f(x)$. Thus, $F(x)$ is strictly increasing if

$$b > a \implies F(b) - F(a) = \int_a^b f(x)dx > 0.$$

We see that a sufficient condition is that $f(x) > 0$ everywhere. Then, we are integrating (or summing) strictly positive numbers, which must result in a strictly positive number. However, note that it is okay for $f(x)$ to equal zero on a subset of $\mathbb{R}$ “of measure zero”. Stated differently, we need $f(x) > 0$ only “almost always”. For instance, $F(x) = x^3$ is clearly a strictly increasing function. Indeed, $f(x) = 3x^2 > 0$ when $x \neq 0$, yet it holds that $f(0) = 0$.

In short, the following statements are technically speaking *not* equivalent:

- (i) $F(x)$ is continuously differentiable and strictly increasing.
- (ii) $F(x)$ is continuously differentiable and $F'(x) = f(x) > 0$ for all x .

The second statement is stronger and implies the first.

Similarly, a function can be strictly convex even if the second derivative is equal to zero on a set of measure zero. Consider, for example, the function $F(x) = x^4$.

3 Properties and useful techniques

There are many useful properties and techniques that turn out to be useful when dealing with integrals. A selection follows.

1. Addition of integrals: If $a < b < c$,

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx.$$

By the way, notation is often abused here. If the domain of $f(x)$ is $[a, c]$ then when we are integrating over the entire domain (as on the LHS) it is common to just write $\int f(x)dx$. If we are not integrating over the entire domain, then we need to clarify what the interval is (as on the RHS).

2. Reversing the direction of integration changes the sign of the integral,

$$\int_a^b f(x)dx = - \int_b^a f(x)dx.$$

3. Integration by parts: Consider the product of two functions $G(x)f(x)$ with $g(x) = G'(x)$ and $f(x) = F'(x)$. An extremely useful rule is then that

$$\begin{aligned}\int_a^b G(x)f(x)dx &= [G(x)F(x)]_a^b - \int_a^b g(x)F(x)dx \\ &= G(b)F(b) - G(a)F(a) - \int_a^b g(x)F(x)dx.\end{aligned}$$

4. Integration by substitution/change of variables: Consider two continuously differentiable functions, f and g . Imagine you want to evaluate an integral of the form

$$\int_a^b f(g(x))g'(x)dx.$$

Instead of thinking of x as the variable, we can think of $u = g(x)$ as the variable. Note that the term $g'(x)dx$ equals du . Hence, we obtain

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du,$$

where we adjusted the limits of integration to reflect that u takes different values than x .

5. Differentiating an integral with respect to a parameter: Let t denote some parameter. We write $f(x, t)$ and assume that it is differentiable with respect to t . Imagine you wanted to see how the integral

$$\int_{a(t)}^{b(t)} f(x, t)dx$$

depends on the parameter t . In other words, we want to differentiate the integral with respect to t .

Note that t enters the integral in three different places; through $a(t)$, $b(t)$, and $f(x, t)$. Since each of these terms change when t changes, we expect our derivative to have three terms. The (awesome) *Leibniz's rule* confirms this intuition. Specifically, it says that

$$\frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t)dx = f(b(t), t)b'(t) - f(a(t), t)a'(t) + \int_{a(t)}^{b(t)} \frac{\partial f(x, t)}{\partial t} dx.$$

4 Differential equations; the integrability question

Integration is key to solving differential equations. Hence, we are now prepared to consider some basic differential equations. We will do so in the context of the

integrability question discussed in lectures. That is, how do we recover or learn preferences from observed demand?

There are two step: (1) Deriving $e(p, u)$ from $x(p, w)$ and (2) using $e(p, u)$ to derive preferences. The first step involves integration, thus explaining why the problem is termed the integrability question. The general approach and some intuition is outlined in the Appendix. For now, we focus on a simple example.

Example: Assume that the utility function is the most basic Cobb-Douglas utility functions, $u(x_1, x_2) = x_1 x_2$. However, we look at the problem from the viewpoint of an outside observer who does not know the utility function. Marshallian demands are

$$\begin{aligned} x_1(p, w) &= \frac{w}{2p_1} \\ x_2(p, w) &= \frac{w}{2p_2}. \end{aligned}$$

We can conceivably “trace out” the Marshallian demands if we observe the agent for long enough. In this sense, they are observable.

Step 1: We will start by deriving the expenditure function using only Marshallian demand (but not the utility function). To do so, we need a differential equation and an “initial condition”. For the differential equation, we use Shephard’s lemma and the link between Marshallian and Hicksian demand to conclude that

$$\frac{\partial e(p, u)}{\partial p_1} = h_1(p, u) = x_1(p, e(p, u)), \quad (3)$$

Note that we are considering the change in only one price (the appendix discusses the more general case). Thus, we have an ordinary differential equation, in which we are trying to infer the function $e(p, u)$ and in which the independent variable is p_1 .

For the initial condition, let us fix some initial economic environment, (p^0, w^0) (known to the outside observer), where we observe consumption $x_1 = \frac{w^0}{2p_1^0}$ and $x_2 = \frac{w^0}{2p_2^0}$. Remember that we pretend not to know the utility function, so let us just refer to utility in this instance as u^0 . Recall that (given non-satiation),

$$e(p^0, u^0) = w^0. \quad (4)$$

This is our initial condition, as it tells us what the expenditure function is in the initial economic environment.

We now consider the following thought-experiment: Income, w^0 , and the price of good 2, p_2^0 , stay fixed, but we observe changes in the price of p_1 , and observe how demand change. Thus, we think of $e(p, u)$ as a function only of p_1 . We will not use (3) and (4) to derive $e(p, u)$.

As the price changes, we know from (3) that

$$\frac{\partial e(p, u^0)}{\partial p_1} = x_1(p, e(p, u^0)) = \frac{e(p, u^0)}{2p_1}. \quad (5)$$

This tells us (given that we only change p_1) that

$$\frac{de}{dp_1} = \frac{e}{2p_1} \iff \frac{de}{e} = \frac{1}{2} \frac{dp_1}{p_1}.$$

We now integrate – or find the antiderivative – on both sides

$$\ln e + c = \frac{1}{2} \ln p_1 + c'.$$

You can check that we did this correctly by differentiating both sides and then note that we are back where we started. We can now rewrite¹

$$\begin{aligned} \ln e &= \ln (p_1)^{\frac{1}{2}} + c' - c \\ &= \ln \sqrt{p_1} + \ln \exp(c' - c) \text{ where } \exp(x) \text{ is Euler's number (2.71828) to the } x\text{th power.} \\ &= \ln \sqrt{p_1} + \ln k, \text{ where } k = \exp(c' - c) > 0 \\ &= \ln k \sqrt{p_1}. \end{aligned}$$

Thus, we conclude that the expenditure function takes the form

$$e(p_1, p_2^0, u^0) = k \sqrt{p_1},$$

as a function of p_1 . However, k is still unknown. To find it, we use the initial condition, (4), that

$$e(p^0, u^0) = w^0.$$

Combining the two yields

$$k \sqrt{p_1^0} = w^0 \iff k = \frac{w^0}{\sqrt{p_1^0}}.$$

In conclusion,

$$e(p_1, p_2^0, u^0) = \frac{w^0}{\sqrt{p_1^0}} \sqrt{p_1} = w^0 \sqrt{\frac{p_1}{p_1^0}}.$$

Let us confirm that this conclusion is correct. Given the utility function $u(x_1, x_2) = x_1 x_2$, we can derive the expenditure function from the expenditure minimization problem (this is an easy exercise), and verify that

$$e(p, u) = 2\sqrt{up_1 p_2}.$$

From utility maximization, we know that indirect utility at (p^0, w^0) is

$$u^0 = \frac{w^0}{2p_1^0} \frac{w^0}{2p_2^0}.$$

¹We usually use simpler notation for $\exp(x)$, simply writing e^x . However, we have already used e to denote the expenditure function, so we need different notation to avoid confusion.

Let us now evaluate

$$\begin{aligned} e(p_1, p_2^0, u^0) &= 2\sqrt{\left(\frac{w^0}{2p_1^0} \frac{w^0}{2p_2^0}\right) p_1 p_2^0} \\ &= w^0 \sqrt{\frac{p_1}{p_1^0}}, \end{aligned}$$

which is exactly what we found by integrating observed demand!

Step 2: With the expenditure function in hand, we turn next to the task of recovering preferences, $\succeq$. The intuition is explained in the appendix, so here we will focus on verifying that the method works rather than why. We will consider the boundary of the set

$$\begin{aligned} V &= \{x \in \mathbb{R}_+^2 \mid p_1 x_1 + p_2^0 x_2 \geq e(p_1, p_2^0, u^0) \text{ for all } p_1 > 0\} \\ &= \{x \in \mathbb{R}_+^2 \mid p_1 x_1 + p_2^0 x_2 \geq w^0 \sqrt{\frac{p_1}{p_1^0}} \text{ for all } p_1 > 0\}. \end{aligned}$$

If $x \in V$ then the bundle always costs at least as much as the cheapest bundle that achieves u^0 , *regardless* of what p_1 is. For any given p_1 , the constraint in the set is satisfied if

$$p_2^0 x_2 \geq w^0 \sqrt{\frac{p_1}{p_1^0}} - p_1 x_1,$$

but remember that we want the constraint to be satisfied for all p_1 . The right hand side is concave in p_1 , and maximized at the (interior) p_1 value where

$$\begin{aligned} w^0 \frac{1}{2} \sqrt{\frac{1}{p_1 p_1^0}} - x_1 &= 0 \\ \frac{w^0}{x_1} \frac{1}{2} \sqrt{\frac{1}{p_1^0}} &= \sqrt{p_1}. \\ \left(\frac{w^0}{x_1} \frac{1}{2} \sqrt{\frac{1}{p_1^0}} \right)^2 &= p_1 \end{aligned}$$

This is the p_1 price at which the constraint is hardest to satisfy. Thus, the constraint is satisfied for all p_1 if and only if it is satisfied at this particular value of p_1 . Substituting into the constraint yields

$$\begin{aligned} x_2 &\geq w^0 \sqrt{\frac{1}{p_1^0} \frac{w^0}{x_1} \frac{1}{2} \sqrt{\frac{1}{p_1^0}}} - \left(\frac{w^0}{x_1} \frac{1}{2} \sqrt{\frac{1}{p_1^0}} \right)^2 x_1 = \frac{1}{4} (w^0)^2 \frac{1}{p_1^0} \frac{1}{x_1} \text{ or} \\ x_1 x_2 &\geq \frac{1}{4} (w^0)^2 \frac{1}{p_1^0 p_2^0}, \end{aligned}$$

the boundary of which is given by the (x_1, x_2) bundles that satisfy

$$x_1 x_2 = \frac{(w^0)^2}{4p_1^0 p_2^0}.$$

The claim is that we have now found the indifference curve. Let us verify that this is true. Given the true utility function, $u(x_1, x_2) = x_1 x_2$ and demand in the initial economic environment, $x_1 = \frac{w^0}{2p_1^0}$ and $x_2 = \frac{w^0}{2p_2^0}$, we see that initial utility was

$$u\left(\frac{w^0}{2p_1^0}, \frac{w^0}{2p_2^0}\right) = \frac{w^0}{2p_1^0} \frac{w^0}{2p_2^0} = \frac{(w^0)^2}{4p_1^0 p_2^0},$$

meaning that the indifference curve through this point is indeed given exactly by

$$x_1 x_2 = \frac{(w^0)^2}{4p_1^0 p_2^0}.$$

In other words, we correctly inferred the agent's indifference curve.

Appendix: General explanation of the integrability question

There are two step to answering the integrability question: (1) Deriving $e(p, u)$ from $x(p, w)$ and (2) using $e(p, u)$ to derive preferences.

Step 1 Recovering $e(p, u)$ from $x(p, w)$: Suppose you know $x(p, w)$ but not yet $e(p, u)$. We start from Shephard's lemma,

$$\frac{\partial e(p, u)}{\partial p_i} = h_i(p, u),$$

but this depends on Hicksian, not Marshallian, demand. Nevertheless, we also know that $h_i(p, u) = x_i(p, e(p, u))$. Hence,

$$\frac{\partial e(p, u)}{\partial p_i} = x_i(p, e(p, u)), \quad i = 1, \dots, L. \quad (6)$$

Let us now fix some initial price vector, p^0 , and the associated level of expenditure, w^0 . Let us now invent a number that we will call u^0 , and think of this as the utility that is achieved with bundle $x(p^0, w^0)$. Note that u^0 is arbitrary, because we know that it's absurd to hunt for a unique utility function. By definition, it then holds that

$$e(p^0, u^0) = w^0, \quad (7)$$

so we know what the expenditure function is for parameters (p^0, u^0) . Let us now hold fixed u^0 and ask if we can derive $e(p, u^0)$ for all possible p . But this is conceptually easy (in practice, it may be extremely hard). Just combine (6) and (7),

$$\begin{aligned}\frac{\partial e(p, u^0)}{\partial p_i} &= x_i(p, e(p, u^0)), \quad i = 1, \dots, L \\ e(p^0, u^0) &= w^0,\end{aligned}$$

and note that this is “simply” a system of partial differential equations with an initial condition. Under an “appropriate condition”, this can be solved for the function $e(p, u^0)$, and we are done! You can now see why this is called the integrability problem; we are solving this system of differentiable equations, i.e. essentially integrating to get $e(p, u^0)$. It turns out that the “appropriate condition” is exactly that the Slutsky matrix (i.e. the matrix of derivatives of $\frac{\partial e(p, u^0)}{\partial p_i}$) is symmetric. This confirms the importance of the symmetry property.

Step 2 recovering preferences from $e(p, u)$: Suppose you know $e(p, u)$, perhaps from the above process. The main idea for recovering preferences is as follows. We just start from the definition of the expenditure function. Given the prices, it is the cheapest way to earn utility u or higher. That is,

$$e(p, u) = \min_x p \cdot x \text{ s.t. } u(x) \geq u.$$

The problem is of course that we don’t know $u(x)$, but would like to find it.

Let us reconsider the constraint again. It says that x should be among the set of bundles that give utility of at least u . Right now, this set is unknown, since we don’t yet know preferences. Nevertheless, let’s call this set V_u . Note that V_u is essentially the superior set to any bundle that gives utility u . However, this is exactly what we are after (remember the preference relation $\succsim$ is defined precisely in terms of which bundles are better; alternatively, the boundary of V_u must be the indifference curve). So we are done if we can just find V_u .

In the reformulation of the problem,

$$e(p, u) = \min_x p \cdot x \text{ s.t. } x \in V_u.$$

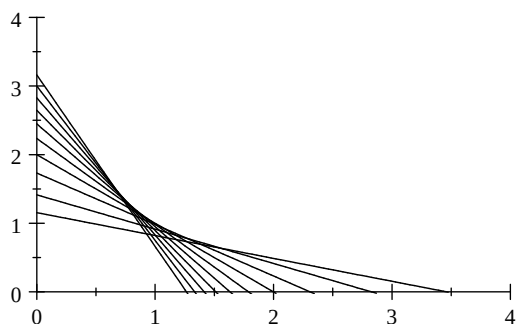
Let us make an educated guess on what V_u might look like and then explain the intuition [MWG presents a different and more detailed argument, including a more thorough examination of the relevant duality theory]. The guess is that

$$V_u = \{x \in \mathbb{R}_+^L | p \cdot x \geq e(p, u) \text{ for all } p \gg 0\}.$$

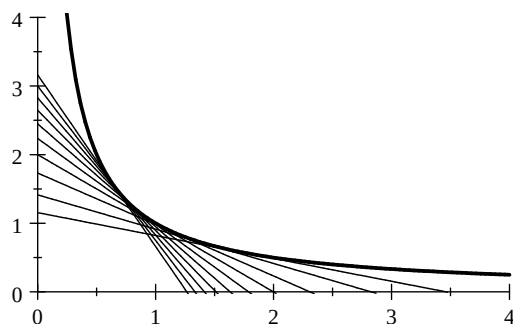
Intuitively, if $x \in V_u$ then it always costs at least as much as the cheapest bundle that achieves u , *regardless* of what the prices are.

The argument then is that $x \in V_u$ must give utility of at least u . To explain, it might be easiest to consider the complementary set, i.e. bundles that are outside of V_u . Such bundles are for *some* p strictly cheaper than the bundle that minimizes expenditures while achieving utility u , $p \cdot x < e(p, u)$. Hence, the bundle in question must be a strictly inferior bundle because otherwise the expenditure minimization problem wasn't solved correctly in the first place. This argument does not apply to bundles in V_u ; those bundles might not be chosen simply because they were too expensive.

It is helpful to give a graphical account of what is going on. As before, fix a utility level u^0 . For any price vector, p , trace out the set of bundles for which $p \cdot x = e(p, u^0)$. Now do this for all price vectors p . As you add more and more of these budget lines, you end up tracing out a curve – the “upper envelope” of the budget lines. See panel (a) in the figures below. This is precisely the indifference curve; see panel (b).² This makes sense because we know that (when preferences are well-behaved) the solution to the expenditure minimization problem is where the indifference curve is tangent to the budget line, so all we are really doing is working our way backwards.



(a) $p \cdot x = e(p, u^0)$ for many p



(b) Superimposing the indifference curve

²The figures are for the case of a Cobb-Douglas utility function, $u(x_1, x_2) = x_1 x_2$, with $u_0 = 1$. In this case, $e(p, u) = 2\sqrt{p_1 p_2 u}$, and $p \cdot x = e(p, u^0)$ simplifies to $x_2 = 2\sqrt{\frac{p_1}{p_2}} - \frac{p_1}{p_2} x_1$. Panel (a) depicts this line for several $\frac{p_1}{p_2}$ values. Panel (b) adds the indifference curve $x_1 x_2 = u^0$.